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Monitoring Electrolyte State of Health in Lithium-Ion Batteries Using Quartz Crystal Microbalance (QCM)

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Master's Programme in Computer and Information Engineering

Abstract

Quartz Crystal Microbalance (QCM) sensors are highly sensitive devices capable of detecting small changes in mass and viscoelastic properties at a surface. Because of this sensitivity, QCM measurements have potential applications for studying electrolyte-related phenomena in lithium-ion battery systems. However, many established QCM setups rely on complex and specialised laboratory instrumentation, which can limit their use in compact or embedded battery monitoring systems.

This thesis investigates signal-processing approaches for analysing QCM sensor signals using a Red Pitaya FPGA-based measurement platform. Two fast QCM measurement approaches, Multi-frequency lock-in amplification (MLA) and the ring-down method, were implemented and evaluated. A modular pipeline combining FPGA-based data acquisition and Python-based post-processing was developed to assess signal quality, stability, and feasibility of these methods.

Experimental data obtained from a commercial QCM-D instrument were analysed to illustrate typical sensor behaviour in an electrolyte environment. The results show that FPGA-based pipelines can capture and analyse key QCM signal characteristics, although hardware limitations and signal stability affect parameter extraction. The developed framework provides a foundation for future work toward compact and cost-effective QCM measurement systems for battery research.

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Acronyms

List of abbreviations used

Acronym	**Meaning**
QCM	Quartz Crystal Microbalance
QCM-D	Quartz Crystal Microbalance with Dissipation monitoring
CV	Cyclic Voltammetry
EQCM-D	Electrochemical Quartz Crystal Microbalance with Dissipation monitoring
MLA	Multi-frequency Lock-in Amplification
FPGA	Field Programmable Gate Array
FFT	Fast Fourier Transform
ADC	Analog-to-Digital Converter
DMA	Direct Memory Access
AXI	Advanced extensible Interface
MMIO	Memory-Mapped Input/Output
SEI	Solid Electrolyte Interphase

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1 Introduction

Quartz Crystal Microbalance (QCM) sensors are precision measurement devices that operate based on the resonance behaviour of a piezoelectric quartz crystal. When an alternating electric field is applied, the crystal oscillates at a resonance frequency. Changes at the crystal surface, such as added mass or altered mechanical properties, influence this resonance behaviour and can therefore be detected through shifts in frequency and damping characteristics [1].

Due to high sensitivity, QCM sensors have been explored for use in liquid environments, where changes in material properties occur at interfaces. In recent years, this has led to growing interest in applying QCM based techniques to electrolyte- related studies in lithium-ion battery systems. Variations in electrolyte composition, viscosity, or interfacial behaviour can influence the measured QCM sensor response, making the sensor a potential tool for indirect observation of battery- related phenomena [1].

Previous research has demonstrated that QCM-based measurements can provide insight into electrochemical processes occurring in lithium-ion battery systems. In such studies, both resonance frequency and energy dissipation are monitored in order to obtain information about mass changes and mechanical properties of layers forming at electrode surfaces. These measurements have been used to investigate interphase formation and electrolyte interactions in battery environments [5].

Traditional QCM measurements are commonly performed using frequency-sweep-based approaches, where the resonance characteristics are obtained by scanning a range of excitation frequencies. While this method is effective in many applications, it can be restrictive when applied to systems that exhibit time-varying or transient behaviour. Frequency-Sweep methods can be limited in temporal resolution when studying dynamic systems [2].

To address these limitations, alternative measurement strategies have been proposed that allow faster extraction of relevant information from the QCM response. Among these approaches are transient ring-down techniques and discrete excitation methods such as Multi-frequency lock-in amplification (MLA). While ring-down methods analyse the decay behaviour of the resonator after excitation is stopped, MLA uses a controlled excitation signal often consisting of multiple frequencies applied simultaneously, rather than sweeping continuously, to evaluate

the sensor response. These approaches aim to reduce measurement time and enable more efficient signal analysis [3], [4].

In addition to the measurement techniques themselves, the implementation platform also plays an important role in practical QCM measurement systems. Commercial QCM-D instruments used in laboratory environments are often complex and expensive, which can limit their integration into compact experimental setups or embedded sensing systems. In battery research, these systems are typically used to study interfacial processes under controlled laboratory conditions.

This thesis investigates signal-processing approaches for analysing QCM sensor signals using an FPGA based measurement framework implemented on the Red Pitaya platform. Experimental QCM data obtained from a commercial QCM-D instrument are used as reference measurements to illustrate the typical signal behaviour observed when a QCM sensor interacts with an electrolyte environment relevant to lithium-ion battery systems. The work focuses on the development of structured signal acquisition and analysis pipelines, including implementations of Multi-frequency lock-in amplification (MLA) and ring-down methods, in order to evaluate their feasibility for analysing QCM signals in a simplified and more cost-effective measurement setup.

2 Background

2.1 Quartz Crystal Microbalance (QCM) Basics

The Quartz Crystal Microbalance (QCM) is a sensitive measurement technique that uses the piezoelectric properties of quartz crystals. When an alternating electrical voltage is applied across a quartz crystal, the crystal oscillates mechanically at a specific resonance frequency. This oscillation behaviour forms the basis of QCM sensing.

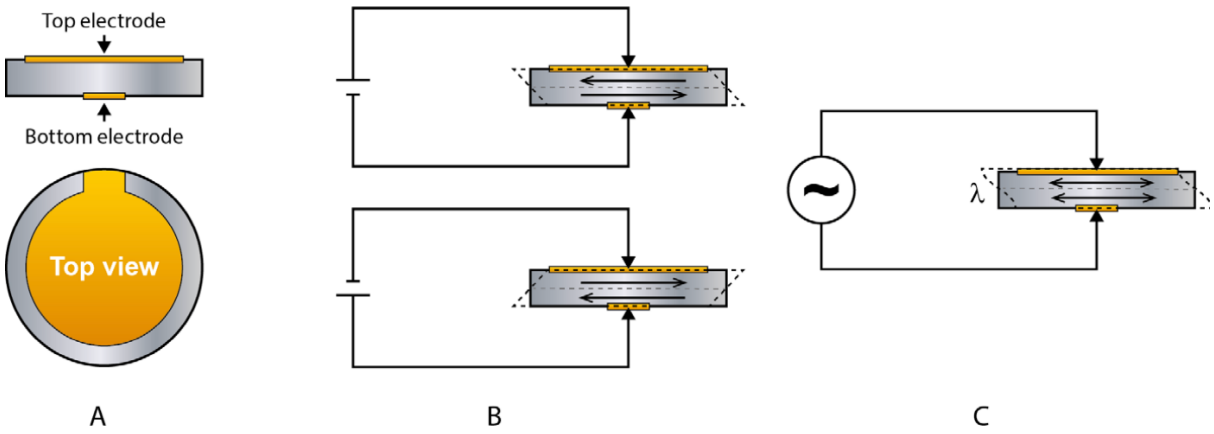


Figure 2.1 ***Basic structure and operating principle of QCM.***

A quartz crystal disk is placed between two metal electrodes (A). When a voltage is applied across the electrodes, the piezoelectricity quartz crystal undergoes mechanical deformation (B). When an alternating voltage is applied, the crystal oscillates in a shear mode parallel to surface (C), forming the basis of QCM sensing. [Adapted from Biolin Scientific, The working principles of QCM and QCM-D technology [6]]

One important characteristic of a QCM sensor is that its resonance frequency changes when mass is added to or removed from the crystal surface. Even very small changes in mass can produce measurable shifts in the resonance frequency [1]. Because of this high sensitivity, QCM sensors are widely used to study surface interactions and thin-film processes.

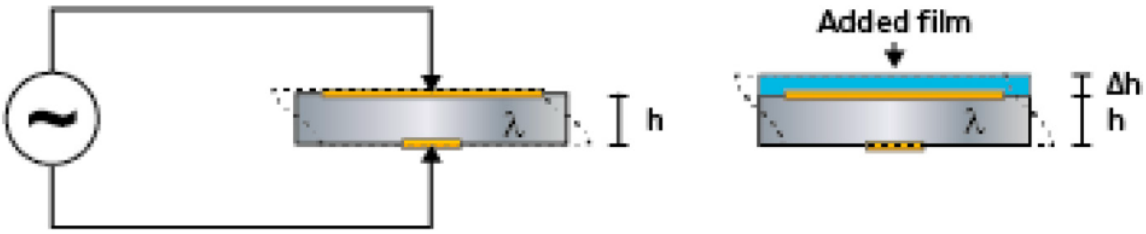


Figure 2.2: ***Conceptual illustration of the QCM mass-loading principle****.*

A quartz crystal oscillates at its resonance frequency when driven by an alternating voltage (left). When an additional film or material is deposited on the crystal surface (right), the effective mass of the oscillating system increases. This increase in mass leads to a decrease in the resonance frequency of the crystal, which forms the basis of QCM sensing. (Adapted from Sauerbrey [1]).

In addition to mass changes, the QCM response is also affected by the mechanical properties of materials interacting with the crystal surface. When the sensor operates in liquid environments, the surrounding medium introduces viscoelastic loading, which affects both the resonance frequency and the energy dissipation of the oscillation. In liquid environments, the damping of the oscillation increases and the signal becomes more complex to analyse [3].

Modern QCM-D systems therefore measure two parameters simultaneously**:** the frequency shift and the dissipation shift. While frequency changes are mainly related to mass changes at the surface, dissipation provides information about the mechanical or viscoelastic properties of the interacting layer.

Another important characteristic of QCM sensors is that the crystal can oscillate not only at its fundamental resonance frequency but also at higher harmonic frequencies known as **overtones**. These overtones correspond to odd multiples of the fundamental frequency, such as the 3rd, 5th, and 7th harmonics. Analysing multiple overtones provides additional information about how the interacting layer behaves mechanically.

Because of these capabilities, QCM techniques have also been applied in electrochemical systems such as **lithium-ion batteries**. In battery systems, processes occurring at the electrode–electrolyte interface can lead to changes in mass and mechanical properties at the surface. Monitoring frequency and dissipation changes can therefore provide insight into electrolyte interactions and interphase formation in battery environments [5]. However, performing reliable QCM measurements in liquid systems can be challenging. The presence of damping, noise, and complex interactions at the interface can make signal interpretation more difficult.

2.2 Conventional QCM Measurement Approaches

Traditional QCM measurements are commonly performed using frequency-sweep techniques. In these methods, the excitation frequency is scanned over a range of values around the resonance frequency of the crystal. From the resulting response curve, the resonance frequency and dissipation characteristics can be determined.

Although this approach is reliable and widely used in commercial QCM-D instruments, it may become less suitable for systems where rapid changes occur. Frequency sweep measurements require scanning many frequency points, which can limit the temporal resolution when studying dynamic processes [2].

In addition, commercial QCM measurement systems are often complex and expensive laboratory instruments. For applications such as battery monitoring or embedded sensing environments, it can be desirable to explore alternative approaches that allow simpler and faster signal acquisition.

These limitations motivate the investigation of measurement techniques that can extract useful information from the sensor response without performing a full frequency sweep.

2.3 Multi-frequency lock-in amplification (MLA)

Multi-frequency lock-in amplification (MLA) is a fast measurement approach designed to analyse QCM signals using a limited number of excitation levels instead of performing a full frequency sweep.

In MLA, the sensor is excited using multiple frequencies applied simultaneously close to the expected resonance frequency of the crystal. By analysing the sensor response at these frequencies, it becomes possible to observe changes in resonance behaviour without scanning the entire frequency range.

The main motivation behind MLA is to reduce measurement time while still capturing relevant information about the resonance response. Since multiple frequencies are evaluated in parallel, the method can provide faster measurements compared to conventional frequency-sweep techniques.

In practical implementations, MLA requires a structured signal-processing pipeline. The recorded signals must first be inspected to ensure that stable oscillation is present and that the signal quality is sufficient for further analysis. Additional checks are typically applied to detect signal clipping, instability, or noise effects.

Multi-frequency lock-in amplification has previously been introduced as a discrete excitation method for fast QCM measurements [4].

In this thesis, MLA is implemented using an FPGA-based acquisition framework on the Red Pitaya platform, combined with Python-based post-processing. The objective is to investigate how MLA signals behave within the developed pipeline and to assess whether the method can be practically applied in such a measurement system.

2.4 Ring – Down Method

Another alternative approach for analysing QCM behaviour is the ring-down method**.** Instead of continuously exciting the crystal, the sensor is first driven near its resonance frequency for a short period. The excitation signal is then turned off, and the natural decay of the oscillation is recorded.

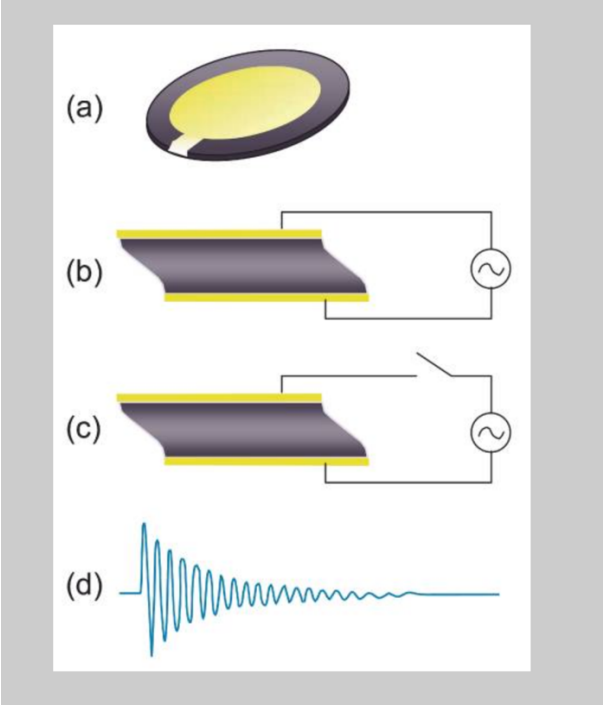


Figure 2.3: ***Conceptual illustration of the QCM ring-down measurement principle****.*

(a) Typical QCM sensor structure with metal electrodes. (b) The quartz crystal oscillates when an alternating electrical signal is applied. (c) When the excitation signal is interrupted, the crystal continues to oscillate freely for a short period. (d) The oscillation amplitude gradually decays over time due to energy losses in the system. Analysis of this decay signal forms the basis of the ring-down method. (Adapted from [2]).

After the excitation is removed, the oscillation amplitude gradually decreases due to energy losses in the system. The shape of this decay contains information about damping and mechanical losses within the resonator. By analysing the decay envelope, it is possible to estimate parameters such as the decay constant or the quality factor of the crystal. Ring-down measurements have been reported in the literature as a method for analysing QCM damping behaviour [3].

Compared to frequency sweep methods, the ring-down approach can potentially provide information within a shorter time window, since it relies on analysing the decay signal rather than scanning frequencies continuously. However, reliable implementation requires precise

control of excitation timing, triggering, and signal acquisition. In practice, the decay signal can be sensitive to hardware limitations, noise, and measurement stability.

In this thesis, a structured signal-processing pipeline was developed to capture and analyse ring-down signals using the Red Pitaya platform. The focus of the implementation is on evaluating the feasibility of observing measurable ring-down behaviour within the developed measurement framework

2.5 Hardware Configuration

The experimental setup used in this thesis consists of a Quartz Crustal Microbalance(QCM) sensor, a Red Pitaya FPGA-based measurement board, and basic analog interfacing components. The goal of the setup was to enable controlled excitation of the QCM crystal and acquisition of the resulting electrical response. The QCM sensor was connected to the Red Pitaya board through a 50 ohm interface to provide approximate impedance matching between the signal source and the crystal. This configuration helps improving signal transmission and reduce signal reflections, supporting stable excitation conditions. The excitation signal was generated directly by the Red Pitaya board.

The Red Pitaya platform was used for both signal generation and signal acquisition. The board allows waveform generation near the resonance frequency of the crystal and provides high-speed analog-to-digital conversion for capturing the response. Depending on the method being tested, acquisition was performed either through FPGA-based configurations or through API-based control.

The overall hardware chain can be summarized as:

Red Pitaya Signal Generator → QCM Sensor → Analog Front-End → Red Pitaya ADC → Data Storage.

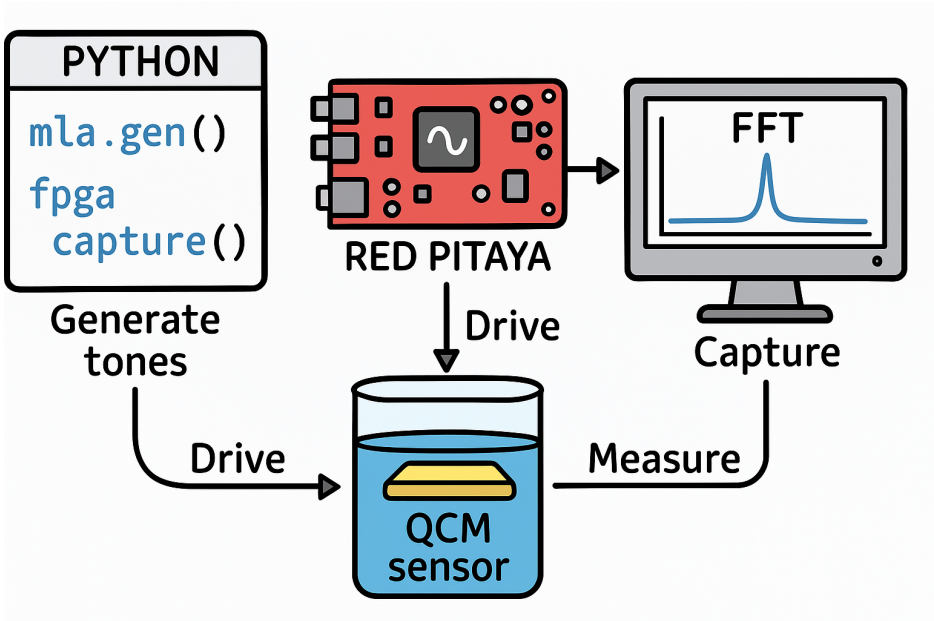


Figure 2.4:Overview of the FPGA-based QCM measurement framework used in this work.

Python scripts are used to generate excitation signals and configure the FPGA-based Red Pitaya platform. The Red Pitaya drives the QCM sensor and captures the resulting electrical response. The recorded signals are then processed using frequency-domain analysis, such as Fast Fourier Transform (FFT), to identify resonance behaviour and analyse the QCM response

The primary objective of this configuration was not to build a finalized sensing system, but to create a flexible test platform for evaluating MLA and ring-down measurement approaches.

2.6 Signal Generation and Acquisition Strategy

For both MLA and ringdown experiments, the QCM crystal was excited near its expected resonance frequency. In the MLA implementation, a multi-frequency excitation signal was applied, and the steady-state response was recorded. In the ring-down implementation, the crystal was excited for a short duration and the excitation was then turned off to observe the natural decay of the oscillation. Data acquisition was performed using the Red Pitaya deep memory, where captured samples are stored in onboard memory (DDR) to enable higher sampling rates and longer recording durations. The acquired signals were exported for further processing in Python, where structured signal-processing pipelines were applied. Special attention was given to trigger configuration, sampling rate selection, and buffer management, as these parameters significantly influence the quality of the captured signals. Incorrect configuration can lead to aliasing, insufficient resolution, or distorted transient behaviour.

2.7 Software and Processing Framework (Python + FPGA Wrapper)

The measurements system is controlled through Python scripts that directly interface with Red Pitaya FPGA registers and DDR memory. Low-level memory-mapped I/O (MMIO) access is used instead of high-level APIs to allow precise control over waveform generation, excitation timing, and data acquisition, which is essential for consistent and repeatable measurements.

The MLA acquisition process follows a structured sequence. The FPGA is first reset and configured, after which the excitation waveform (either frequency comb or a pure tone) is generated and written into DDR memory. The FPGA then controls the timing of signal generation and acquisition, and the captured ADC data is stored in memory and later retrieved for processing.

This structured approach ensures that each stage of the measurement is explicitly controlled, which is important because FPGA-based acquisition is sensitive to timing, buffering, and synchronization issues. Errors in these steps can lead to incorrect or unstable measurements.

For ring-down measurements, a similar acquisition approach is used, with modifications to the excitation timing so that the decay signal after excitation can be captured. This enables analysis of the transient response of the QCM sensor.
The system operates at a fixed sampling rate of 125 MHz, determined by the Red Pitaya hardware. This sampling rate is used for both MLA and ring-down measurements and provides sufficient temporal resolution to capture the high-frequency oscillations of the QCM signal.

Data acquisition is performed using onboard DDR memory (deep memory mode), where captured samples are stored for later retrieval. This allows longer recording durations at high sampling rates, which would not be possible using limited internal buffers alone.

The acquired data is then processed in Python using a modular signal-processing pipeline. This includes time-domain inspection, FFT-based frequency analysis, and validation steps such as stability checks, noise evaluation, and signal quality assessment.
Separating data acquisition from post-processing allows the raw data to remain unchanged while different analysis approaches are tested. This also makes it easier to identify whether

unexpected behaviour originates from hardware configuration, signal acquisition, or processing steps.

Overall, the combined FPGA and Python framework provides a controlled and flexible environment for evaluating MLA and ring-down measurement approaches in a consistent and repeatable manner.

3 Signal Processing Pipeline Overview

This chapter presents the developed signal-processing pipelines and illustrates their behaviour using FPGA-acquired data and simulation necessary. The figures are used to demonstrate how the signal evolves through each stage of processing. A detailed evaluation of the performance of the MLA and ring-down methods, including their limitations, is presented in next chapter.

3.1 Overview of the MLA Pipeline

The MLA signal-processing pipeline was developed to transform raw QCM signal data into structured spectral information. Instead of extracting only a single resonance value, the pipeline was designed to evaluate signal quality at multiple stages before interpreting the final response. The acquisition process produces a raw time-domain signal consisting of digitized voltage samples recorded by the Red Pitaya ADC, These samples are stored in binary format together with metadata describing sampling rate, excitation frequencies, and acquisition parameters. The pipeline processes this data in a sequence of modular steps, each responsible for validating a specific property of the signal.

The processing stages can be grouped into three main categories:

- Time-domain validation
- Frequency-domain analysis
- Stability and comb verification

The time-domain stages ensures that the recorded waveform is free from clipping, excessive noise, or DC offset. The frequency-domain stage applies windowing and Fast Fourier Transform(FFT) analysis to identify resonance peaks and estimate signal-to-noise ratio (SNR). The latter stages evaluate frame-to-frame stability and verify that the MLA excitation tones appear at the expected frequency bins.

This structured approach ensures that resonance estimation is not performed blindly. Each stage acts as a validation layer, reducing the risk of misinterpreting distorted or unstable data

3.1.1 Time-Domain Validation (TD0)

Before performing any spectral analysis, the acquired signal must be inspected in the time domain. The raw ADC samples represent the instantaneous voltage response of the QCM crystal. At this stage , the goal is not to extract resonance information, but to verify that the signal is physically meaningful and not corrupted by hardware limitations.

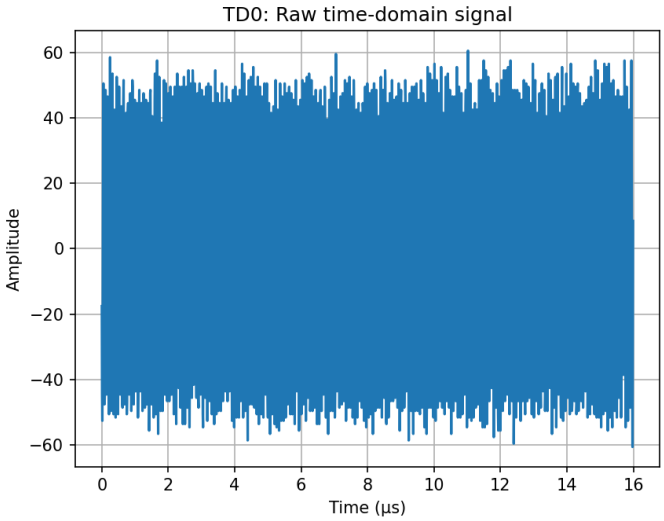


Figure 3.1: Raw time-domain signal captured during the TD0 stage of the MLA pipeline. The plot shows the full acquired waveform from the Red Pitaya measurement system before any signal processing or filtering is applied. This stage is used to inspect the overall signal behaviour and verify that oscillations are present.

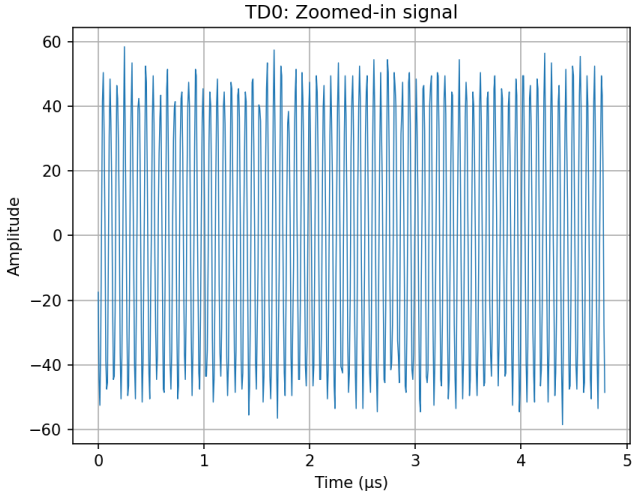


Figure 3.2: Zoomed-in view of the TD0 time-domain signal. The magnified segment highlights the oscillatory behaviour of the captured waveform, allowing verification of signal stability and amplitude consistency prior to further MLA processing.

The first validation consists of plotting the raw waveform over time. This allows visual inspection of:

- Oscillation presence
- Clipping or saturation
- Unexpected amplitude jumps

The ADC operates at a fixed sampling rate of

$$f_s = 125\text{MHz}$$

The time resolution of the recorded waveform is therefore

$$\text{Delta t} = \frac{1}{f_s}$$

For a record containing N samples, the total capture duration is

$$\text{T} = \frac{N}{f_s}$$

These parameters define the observable time window of the signal.

If the amplitude exceeds the ADC dynamic range, clipping occurs. Clipping introduces harmonic distortion in the frequency domain and can invalidate subsequent

FFT-based analysis. Therefore, amplitude verification is performed prior to spectral processing.

Time-domain inspection also allows detection of unstable excitation or incorrect timing between waveform generation and acquisition. Since the MLA method relies on steady oscillatory behaviour, the time-domain waveform must show a stable sinusoidal pattern before proceeding to frequency-domain analysis.

Only after the signal passes these checks is it forwarded to the FFT stage.

3.1.2 Frequency-Domain Analysis(A3-A5)

After validating the signal in the time domain, the next step is to analyse its frequency content. Since the QCM response is oscillatory, resonance information is more clearly observed in the frequency domain.

The conversion from time-domain samples to frequency-domain representation is performed using the Fast Fourier Transform (FFT). Since the acquired MLA waveform is real-valued , a one-sided FFT is computed, representing frequencies from 0 Hz up to the Nyquist frequency $\frac{f_s}{2}$. For a discrete signal consisting of N samples acquired at sampling frequency f_s, the frequency resolution of the FFT is given by:

$$\Delta f = \frac{f_s}{N}$$

This equation shows that increasing the number of samples improves frequency resolution. In the implemented system, the sampling frequency is fixed at:

$$f_s = 125\ MHz$$

Therefore, the achievable frequency resolution depends directly on the chosen record length N.

3.1.2.1 Window Before FFT

Before applying the FFT, a window function is applied to the time-domain signal. This is necessary because the FFT assumes that the signal is periodic with in the captured frame. If the captured waveform does not contain an exact integer number of oscillation periods , discontinuities appear at the edges of the record. These discontinuities cause spectral leakage, where energy spreads into neighbouring frequency bins.

To reduce spectral leakage, a Hanning window in NumPy implementation is applied:

$$x_w[n] = x[n] \,.\, w[n]$$

Where $w[n]$ is the Hanning window in NumPy implementation defined as :

$$w[n] = 0.5\ (1 - \cos(\frac{2\pi n}{N-1}))$$

The purpose of windowing is not to change the signal, but to smooth the edges of the sampled frame so that the FFT more accurately represents the true frequency content .

In the MLA pipeline, windowing ensures that the resonance peak remains well-defined and that peak detection is not affected by the artificial side lobes caused by leakage.

3.1.2.2 Peak Detection and Resonance Identification(A3)

Once the FFT magnitude spectrum is computed, the dominant peak near the expected resonance frequency is identified. In the implemented system, the FFT magnitude is

normalized by dividing by $N/2$, so that the spectral amplitude remains consistent with the signal amplitude. This is performed by locating the maximum magnitude within a defined frequency region of interest.

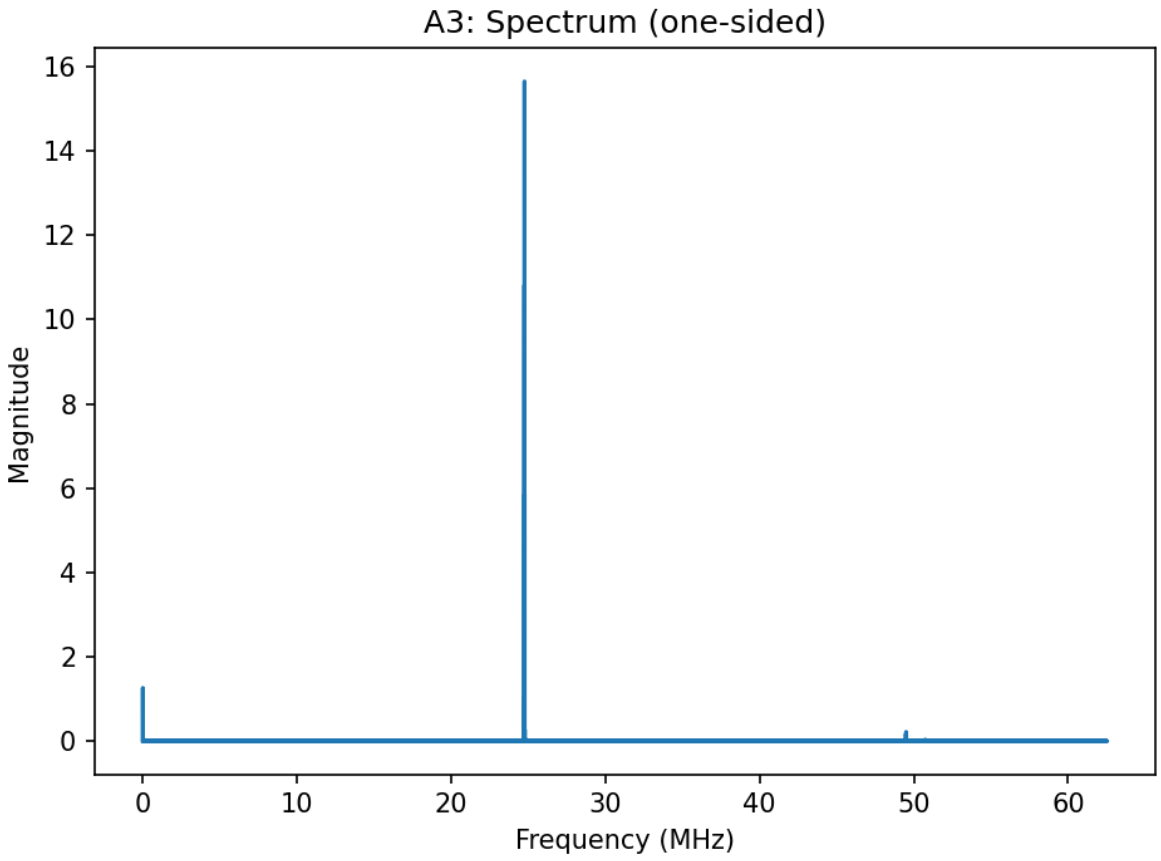


Figure 3.3: Frequency-domain representation of the measured signal during the A3 stage of the MLA pipeline. The spectrum shows a dominant resonance peak near 24.75 MHz, corresponding to the oscillation frequency of the QCM crystal. This step verifies that the acquired signal contains a clear resonance component before further analysis.

Let $X[k]$ represent the FFT magnitude at bin k. The resonance frequency estimate is obtained as:

$$f_{peak} = k_{max} . \Delta f$$

Where k_{max} is the index corresponding to the maximum magnitude.

This step verifies that the crystal is oscillating near the intended frequency and provides the basis for further stability and reliability checks.

3.1.3 Stability and Frequency Tracking (A4)

After identifying the dominant resonance peak in a single FFT frame (A3), it is necessary to verify that the detected frequency remains stable over time. A single peak is not sufficient to confirm reliable oscillation, since transient noise or hardware artifacts may produce temporary maxima.

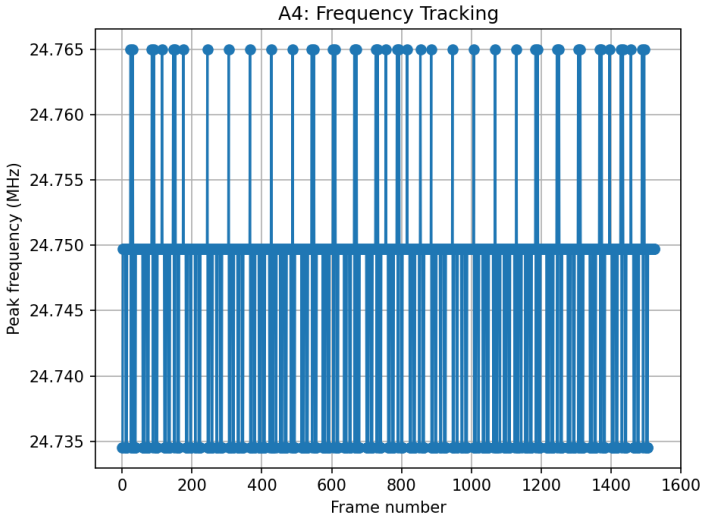


Figure 3.4: Frequency tracking of the detected resonance peak across measurement frames during the A4 stage of the MLA pipeline. This step verifies the stability of the resonance frequency prior to further processing.

To evaluate stability, the acquired signal is divided into multiple consecutive frames. In the implemented system, each frame consist of 8192 samples. An FFT is computed for every frame independently , and the peak frequency is extracted for each segment.

Let the peak frequency detected in frame i be.

$$f_i$$

The frame-to-frame frequency variation is the defined as:

$$\Delta f_i = f_i - f_{i-1}$$

If the oscillation is stable, the variation Δf_i should remain small across frames.

Large fluctuations may indicate:

- Excitation instability
- Timing Misalignment

- Sampling configuration issues
- Hardware- related noise

The overall stability can be evaluated by computing the standard deviation of the detected frequencies:

In the implemented A4 module, this standard deviation is computed directly from the list of detected peak frequencies using a population standard deviation formula.

$$\sigma_f = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(f_i - \overline{f})^2}$$

Where:

- M is the number of frames
- $\overline{f}$ is the mean detected frequency

A low value of σ_f indicates consistent resonance behaviour. A high value suggests instability, making subsequent interpretation unreliable.

This step ensures that resonance identification is not based on a single snapshot but on consistent behaviour over time.

3.1.4 Signal-to-Noise Ration Estimation(A5)

To ensure that the detected peak is meaningful, the signal-to-noise ratio (SNR) is estimated. A simplified SNR metric is computed as:

where:

$$P_{signal} = \frac{1}{K}\sum|X[k]|^2 \quad \text{(around peak)}$$

$$P_{noise} = \frac{1}{L}\sum|X[k]|^2 \quad \text{(noise region)}$$

$$SNR_{linear} = \frac{P_{signal}}{P_{noise}}$$

$$SNR_{linear} = 10\,log_{10}(SNR_{linear})$$

- P_{signal} is the mean squared FFT magnitude computed over several bins around the detected resonance peak.
- P_{noise} is the mean squared FFT magnitude computed from bins located sufficiently far from the resonance frequency.

In the implemented , a ±1 MHz region around the resonance peak is excluded from the noise calculation to prevent contamination from the resonance skirt. This provides a relative measure of peak isolation.

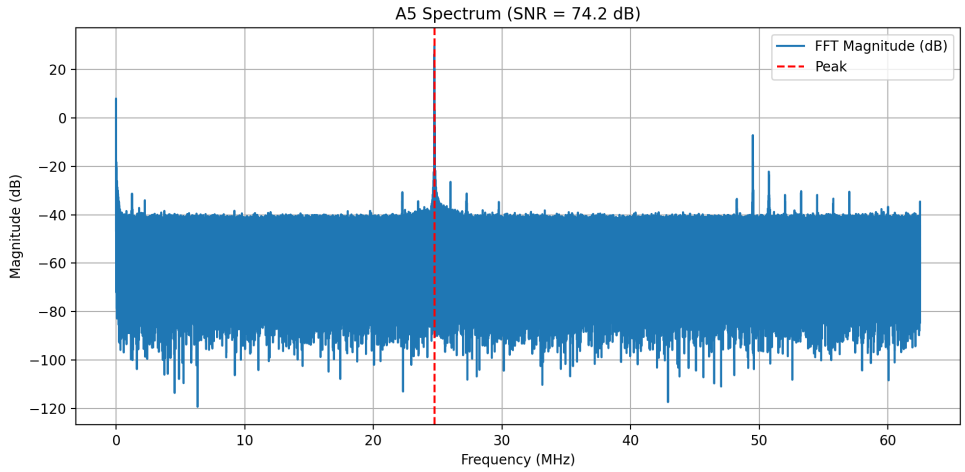


Figure 3.5: Frequency spectrum obtained during the A5 stage of the MLA pipeline. The dominant resonance peak is clearly visible above the noise floor, resulting in a signal-to-noise ratio (SNR) of approximately 74.2 dB. This confirms that the captured signal contains a sufficiently strong resonance component for reliable MLA analysis.

In the implemented A5 module, multiple bins around the peak are averaged in order to reduce sensitivity to bin misalignment and small frequency shifts.

A low SNR indicates that the resonance is not clearly distinguishable from noise, and further analysis may be unreliable.

3.1.5 Reliability Check (A6)

Even if the resonance peak is identifiable (A3) and appears stable across frames (A4), the underlying time-domain wave must still be verified. Frequency-domain analysis assumes that the measured signal is smooth and physically consistent. If the waveform contains clipping or sudden discontinuities, the FFT spectrum may become distorted.

For this reason, A6 performs a time-domain reliability check before further MLA processing.

First, the DC offset is removed so that the signal is centred around zero:

$$x_{clean}[n] = x[n] - mean\,(x)$$

Two condition are then evaluated.

3.1.5.1 Clipping Detection

Clipping occurs when the signal reaches the ADC limits. After normalization, clipping is detected if the waveform touches its maximum or minimum allowed value. Clipping distorts the sinusoidal shape and introduces artificial frequency components in the FFT spectrum. If clipping is present, the frame is considered unreliable.

3.1.5.2 Smoothness (Step-size) check

A physically realistic QCM oscillation should vary smoothly from one sample to the next. To detect sudden jumps, the sample-to-sample difference is calculated.

$$step[n] = |x[n] - x[n-1]|$$

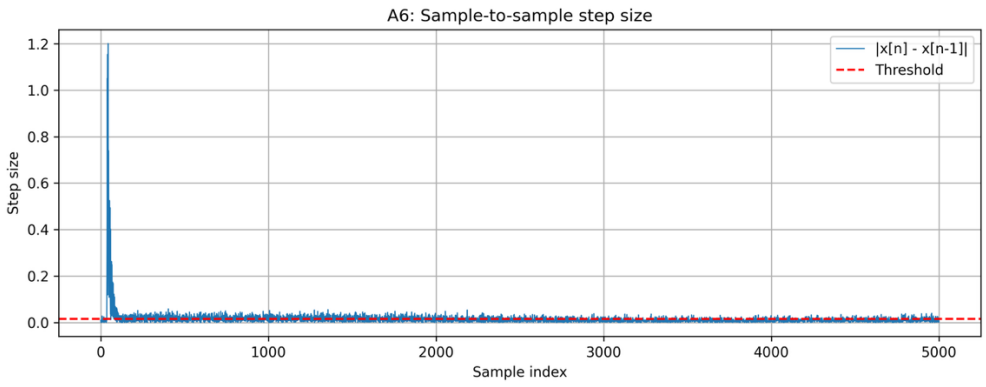


Figure 3.6: Sample-to-sample step size analysis during the A6 stage of the MLA processing pipeline. The absolute difference between consecutive samples is evaluated to detect abrupt signal changes or discontinuities. The red dashed line represents the threshold used to identify abnormal jumps in the signal.

The maximum step size is then evaluated. If this value exceeds a predefined threshold, the wave waveform is flagged as unstable. Such abrupt changes typically indicate electrical noise, grounding issues, or excitation instability rather than genuine physical changes in the crystal.

By combining clipping detection and smoothness evaluation, A6 ensures that the waveform behaves like a stable sinusoidal oscillation before frequency extraction is performed. This prevents distorted peak detection in A3, unstable tracking in A4, and unreliable SNR estimation A5.

3.1.6 Amplitude and Clipping Safely (A7)

After verifying waveform stability A6, the next step is to ensure that the signal amplitude is operating within a safe range. Even a smooth and stable waveform can still cause problems if it is too weak or too close to clipping.

The MLA method assumes that the QCM crystal is driven by a clean sinusoidal signal. If the amplitude is too high, the ADC may saturate and flatten the peaks of the waveform. This distortion introduces artificial frequency components in the FFT spectrum and can affect resonance estimation. On the other hand, if the amplitude is too low, the signal may be comparable to the noise level, reducing the reliability of peak detection.

To evaluate this, the signal is first centred and normalized:

$$x_{norm}[n] = \frac{x[n]}{\max\ |x[n]|}$$

The peak amplitude is then defined as:

$$peak = max|x_{norm}[n]|$$

Clipping is detected if:

$$peak\ \geq 1.0$$

In addition, a predefined safe operating region is used to determine whether the signal is too weak, safely driven or close to saturation. This classification does not modify the signal, it only evaluated whether the chosen excitation amplitude is appropriate.

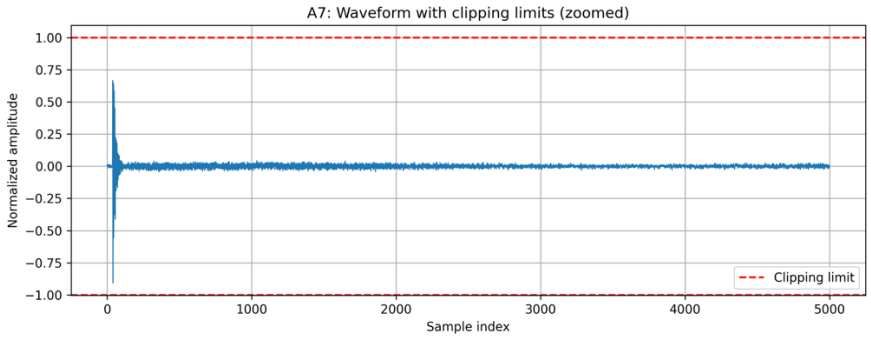


Figure 3.7: Clipping verification performed during the A7 stage of the MLA pipeline. The normalized waveform amplitude is compared against predefined clipping limits (±1) to ensure that the signal remains within the valid dynamic range of the acquisition system. This step confirms that the captured signal is not affected by saturation.

By checking amplitude conditions before further analysis, A7 ensures that the waveform is neither distorted nor excessively weak. This protects the accuracy of SNR estimation (A5) and subsequent resonance interpretation in the MLA pipeline.

3.1.7 Comb-Size and Tone Alignment Check (A8)

The MLA method can evaluate one or multiple tones (overtone frequencies) within the same acquisition frame. For the analysis to be valid, each tone must be correctly aligned with the FFT frequency grid, and the signal around each tone must be strong enough to be reliably measured.

In A8, the FFT bin frequencies are defined as:

$$f_{bin}[k] = k \,.\frac{f_s}{N}$$

Where f_s is the sampling frequency and N is the number of samples.

For each expected tone frequency f_{target}, the closest FFT bin is identified. The alignment error is then calculated as:

$$error = |f_{target} - f_{bin,peak}|$$

If this error is large compare to the FFT bin width, the tone may not be properly represented in the spectrum. In such cases, frequency estimation accuracy may decrease

In addition to alignment, a local signal-to-noise ratio is evaluated around each tone:

$$SNR_{local} = peak_{db} - noise_{db}$$

This value indicates whether the selected tone is clearly distinguishable from the surrounding spectral background.

If only one tone is present, the comb-size representation reduces to a single point. If multiple tones are used, each one is evaluated individually for alignment and signal strength.

By verifying tone alignment and local SNR, this module ensures that the selected MLA tones are correctly positioned and sufficiently strong before further interpretation. This step confirms that the chosen comb-size and bandwidth settings are appropriate for the measurement.

3.1.8 Magnitude and Phase Analysis (A9)

After verifying peak detection (A3), stability (A4), signal strength (A5), reliability (A6), amplitude safety (A7), and tone alignment (A8), the final MLA step provides a combined magnitude and phase view of the signal.

While the previous steps focus mainly on the magnitude spectrum, A9 additionally evaluated the phase response of the FFT. This gives a more complete frequency-domain description of the oscillation.

The FFT of the windowed signal is computed as:

$$X(k) = \sum_{n=0}^{N-1} x[n]w[n]e^{-j2\pi kn/N}$$

This operation converts the time-domain waveform into its frequency components.

From this, the magnitude spectrum is obtained as:

$$|X[k]| = \sqrt{Re(X|k|)^2 + Im(X|k|)^2}$$

The magnitude shows how strong the signal is at each frequency.

The phase spectrum is obtained as:

$$\phi[k] = \angle \mathrm{X[k]}$$

The phase describes the relative timing of each frequency component. Since phase values can mathematically jump between +π and –π, phase unwrapping is applied so that the phase curve appears smooth and continuous.

The resonance frequency is again identified as:

$$f_{peak} = k_{max} \cdot \frac{f_s}{N}$$

This confirms that the dominant frequency observed earlier remains consistent.

For visual inspection, an approximate impedance-style representation is also calculated:

$$Z(F) \approx \frac{1}{|X(f)|}$$

This is not a full electrical impedance model, but a simple way to visualise how the resonance appears when viewed in inverse magnitude form.

The purpose of A9 is mainly diagnostic. It allows visual inspection of:

- The shape of the resonance peak
- The smoothness of the phase response
- Possible distortions or irregularities
- Overall frequency- domain behaviour

By combining magnitude and phase information, A9 provides a final verification that the MLA signal behaves consistently before any physical interpretation is made.

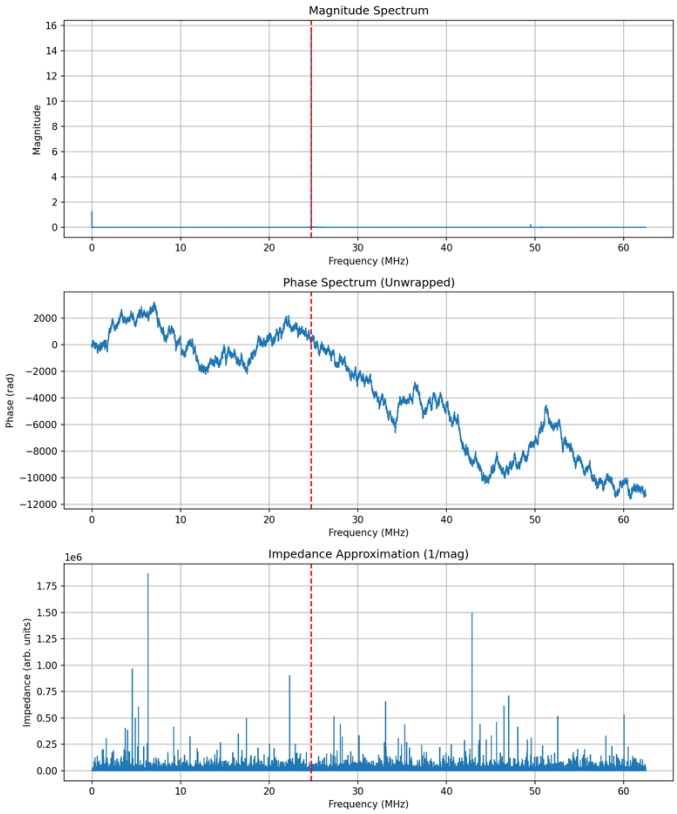


Figure 3.8: Final spectral analysis performed during the A9 stage of the MLA pipeline. The figure presents the magnitude spectrum, unwrapped phase spectrum, and an impedance approximation (1/magnitude) derived from the measured signal. The red dashed line marks the detected resonance frequency of the QCM sensor, providing a comprehensive view of the signal behaviour in the frequency domain.

3.1.9 I/Q Demodulation and Resonance Projection (A10)

While the previous MLA steps analyse the signal using FFT-based inspection , the final step performs direct frequency selective demodulation. This step extracts the response of the QCM at each excitation frequency individually.

Unlike the FFT, which evaluates all frequency components simultaneously, I/Q demodulation processes one selected frequency f_k, two reference signals are generated :

$$\cos(2\pi f_k\, t),\ \sin(2\pi f_k t)$$

The averaging operation acts as a low-pass filter. It removes all the other frequency components and keeps only the response at the selected frequency f_k.

From the in-phase (I) and quadrature (Q) components, the amplitude and phase of the response are calculated as:

$$A_k = \sqrt{{I_k}^2 + {Q_k}^2}$$

$$\phi_k = \tan^{-1}(\frac{Q_k}{I_k})$$

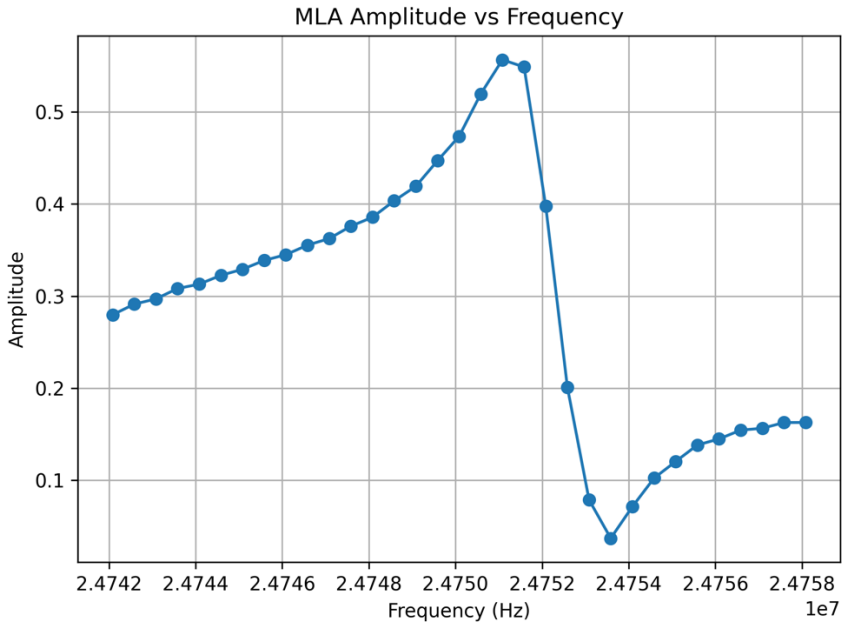


Figure 3.9: MLA amplitude response as a function of frequency around the detected resonance region. The variation in signal amplitude across the discrete excitation frequencies illustrates the resonance behaviour of the QCM crystal captured using the MLA approach.

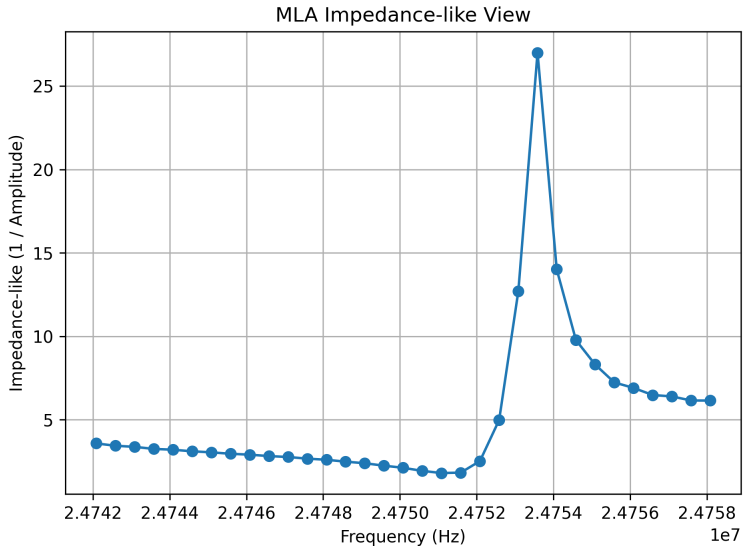


Figure 3.10: Impedance like representation obtained from the MLA analysis around the resonance region of the QCM sensor. The plot shows the inverse amplitude response as a function of frequency, highlighting the resonance peak near 24.75 MHz. This representation provides a visualization similar to the impedance response of the resonator.

The amplitude A_k represents how strong the QCM responds at frequency f_k, while ϕ_k represents the phase shift relative to the excitation.
By repeating this process for all excitation frequencies in the MLA comb, an amplitude-versus -frequency curve is obtained. This curve represents the reconstructed resonance behaviour of the sensor.
For visualisation purposes, an impedance – like representation may also be calculated as:

$$Z_k \propto \frac{1}{A_k}$$

where A_k is obtained from I/Q demodulation of the experimentally measured signal. The figures shown are therefore based on measured data, while the impedance-like curve is a derived quantity computed from the measured amplitude values.

This representation does not correspond to the full electrical impedance model of the QCM, but provides a convenient way to visualise resonance behaviour.

A10 therefore produces the final MLA resonance curve (amplitude and phase versus frequency). This step transforms the validated time-domain signal into physically interpretable frequency-response data.

3.2 Overview of the Ring-Down Pipeline

The Ring-down method analyses the natural decay of the QCM crystal after the excitation signal is removed. Instead of sweeping multiple frequencies, the crystal is excited once with a short burst and its free oscillation is recorded as it gradually decays.

The recorded signal follows an exponentially decaying sinusoidal form:

$$x(t) = A_0 e^{-t/\tau} \sin(2\pi f_0 t + \phi_0)$$

From this decay, two main parameters are extracted:

- The resonance frequency f_0
- The decay constant τ

These are directly related to the quality factor of the resonator:

$$Q = \pi f_0 \tau$$

The Ring-down analysis pipeline used in this work consists of the following processing stages:

- RD0-RD1: Mode selection and extraction of the relax segment
- RD2: Cleaning and normalization of the decay signal
- RD3: Envelope estimation
- RD4: Exponential fitting to extract τ
- RD5: Frequency estimation from zero-crossing
- RD6: Saving and summarizing results
- RD7: Damping and linewidth calculation

Each stage processes the signal step-by-step, transforming the raw decay waveform into physically meaningful parameters that describe the behaviour of the QCM.

To demonstrate the behaviour of the ring-down analysis pipeline, synthetic signals were generated in simulation mode to illustrate the expected transient decay behaviour and the extraction of relevant parameters

3.2.1 Mode selection and Relax Segment Preparation (RD0-RD1)

The first step of the Ring-down pipeline prepares the signal for analysis.
RD0 defines the operating mode and basic parameters such as:

- Sampling frequency f_s
- Expected resonance frequency f_0
- Burst duration
- Relax duration

Only one mode can be selected at a time.

- Simulation Mode
- New File Mode
- Old File Mode

RD1 then generates or load the relax segment depending on the selected mode.
In simulation mode, a synthetic burst followed by an exponential decay is generated :

$$x_{relax}\,(t) = A_0 e^{-t/\tau_{sim}} \sin(2\pi f_0 t\,)$$

In file mode, the relax segment is extracted from the recorded raw data using metadata that specifies the start position and length of the captured region.
The output of RD0 and RD1 is

- Sampling frequency f_s
- The relax signal $x_{relax}\,(t)$

No fitting or frequency estimation is performed at this stage.
This step only prepares the correct portion of the signal for further processing.

3.2.2 Clean and Initial Inspection of the Decay Signal (RD2)

After extracting the relax segment in RD1, the next step prepares the signal analysis.
RD2 prepares the relax signal for analysis by removing offsets and checking whether the recorded decay looks physically correct.

First, the DC offset is removed:

$$x_0[n] = x_{relax}[n] - mean\,(x_{relax})$$

This ensures that the decay oscillates around zero and is not shifted by hardware bias or slicing errors.
Optionally, the signal is normalized:

$$x_{1[n]} = \frac{x_0[n]}{\max\,|x_0[n]|}$$

Normalization does not change the decay behaviour. It only scales the amplitude so that different runs can be compared more easily.

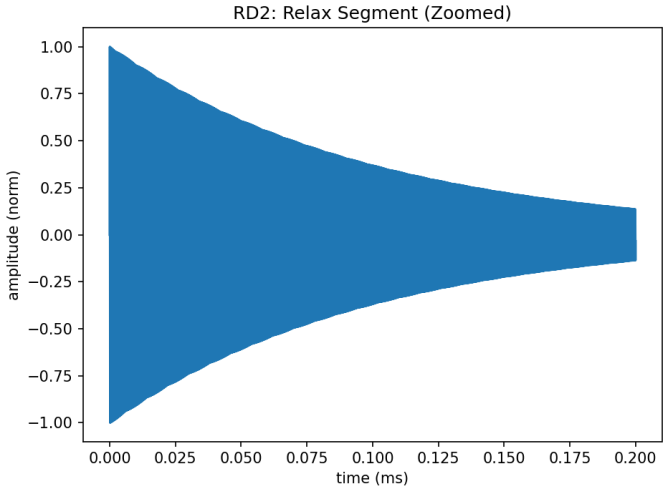


Figure 3.11: Simulated ring-down relaxation signal used to illustrate the RD2 stage of the ring-down analysis pipeline. After excitation, the oscillation amplitude decays exponentially over time as energy dissipates from the resonating QCM crystal. This synthetic signal represents the expected transient behaviour captured during a ring-down measurement.

A short zoomed plot of the beginning of the relax segment is then generated. This allows visual inspection of:

- The Oscillation Shape
- The presence of noise
- Possible Clipping
- Whether the decay behaviour looks reasonable

RD2 does not extract physical parameters.
It ensures that the decay signal is clean and suitable for envelope estimation and exponential fitting in the following steps

3.2.3 Envelope Estimation of the Decay Signal (RD3)

After RD2, the relax signal is centred and optionally normalized.
RD3 converts the oscillating decay signal into smooth amplitude curve that represents how the vibration strength decreases over time.

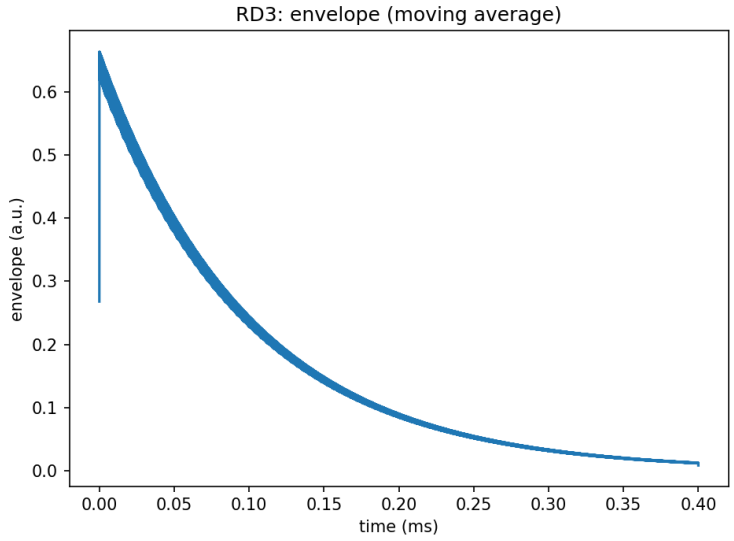


Figure 3.12: Envelope ring-down relaxation signal used to illustrate the RD2 stage of the ring-down analysis pipeline. After excitation, the oscillation amplitude decays exponentially over time as energy dissipates from the resonating QCM crystal. This synthetic signal represents the expected transient behaviour captured during a ring-down measurement.

However, the signal is still moving up and down around zero because the crystal continues to vibrate.

To understand how the vibration becomes weaker with time, the overall decrease in amplitude must be extracted from this waveform.

First, the absolute value of the signal is taken:

$$env_{abs}[n] = |x_1[n]|$$

This removes the positive and negative swings and keeps only the size of the vibration at each point.

Since this curve still contains small variations from the oscillation itself, a moving average is applied to smooth it.
The smoothing window is chosen to be approximately half of one vibration period:

$$M \approx \frac{1}{2}\left(\frac{f_s}{f_0}\right)$$

The envelope is then calculated as :

$$env[n] = \frac{1}{M}\sum_{k=0}^{M-1} env_{abs}\,[n-k]$$

The result is a smooth curve that shows how the amplitude of the vibration decreases over time. This envelope is used in RD4 to fit an exponential decay and determine the time constant τ.

3.2.4 Exponential Fit of the Decay (RD4)

RD4 takes the smooth envelope obtained in RD3 and calculates how quickly the vibration decreases with time.
In other words, this step measures how fast the crystal stops vibrating after the excitation is removed.

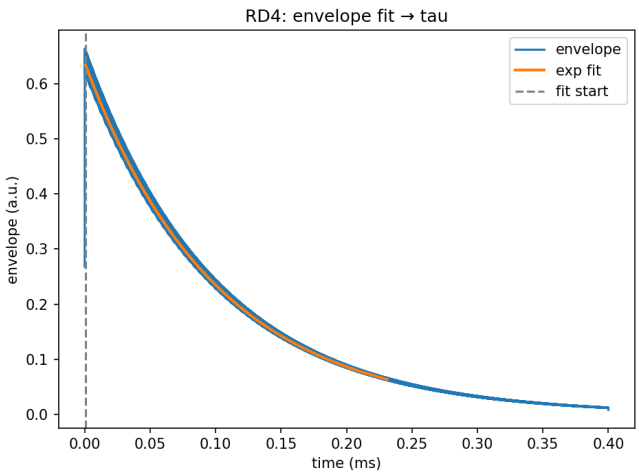


Figure 3.13: Exponential extraction of the simulated ring-down signal during the RD3 stage of the pipeline. A moving-average approach is used to estimate the amplitude envelope of the decaying oscillation, enabling a smoother representation of the transient decay behaviour.

The envelope $E(t)$ is fitted to the exponential model:

$$E(t) = A_0 e^{-t/\tau}$$

Here, τ is the decay time constant.

It tells how long the vibration takes to significantly decrease.

To estimate τ, the envelope is converted into logarithmic form.

$$ln\left(E(t)\right) = \ln\left(A_0\right) - \frac{t}{\tau}$$

A straight-line fit is applied, and from its slope the decay constat τ is obtained.

Once τ is known, the quality factor is calculated:

$$Q = \pi f_0 \tau$$

The quality factor describes how strongly the crystal damped.

- A larger τ means slower decay and higher quality factor Q.
- A smaller τ means faster decay and strong energy loss.

So, RD4 is the step where the decay curve is converted into physical parameters of the QCM.

3.2.5 Estimating the Resonant Frequency from the Decay (RD5)

After RD4 determines how fast the vibration decays, RD5 measures how fast crystal is oscillating during the free decay.

In other words, RD5 calculates the actual resonance frequency f_0 directly from the time signal.

If the time between two consecutive zero-crossing is Δt_{half}, the full period is:

$$T = 2 \, . \, mean \, (\Delta t_{half})$$

The resonance frequency is then:

$$f_0 = \frac{1}{T}$$

To improve accuracy, the exact zero-crossing time is estimated using linear interpolation between two samples.

RD5 therefore provides the resonance frequency directly from the time-domain signal, independent of the exponential fit.

While RD4 measures how fast the vibration decreases,

RD5 measures how fast the crystal oscillates.

Together, they give both key parameters of the Ring-Down method:

- Decay constant τ
- Resonance frequency f_0

3.2.6 Damping Factor and Linewidth Calculation (R6)

After RD4 determines the decay constant τ and RD5 estimates the resonance frequency f_0, RD6 calculates parameters that describes damping.

The first quality is the damping factor:

$$\alpha = \frac{1}{\tau}$$

This value represents how quickly the vibration amplitude decreases.

- A larger α means faster energy decay.

Next, the resonance linewidth is calculated.

The line width describes how broad the resonance would appear in the frequency domain.
From decay constant:

$$\Delta f_\tau = \frac{1}{2\pi\tau}$$

From the quality factor:

$$\Delta f_Q = \frac{f_0}{Q}$$

Both expression describes the same physical effect.
If the calculation are consistent, the two linewidth values should be similar.

RD6 therefore connect the time-domain behaviour to frequency-domain resonance width.
It translates the decay measurements into physical meaningful damping parameters.

3.2.7 Final Summary and Data Logging

RD7 is the final step of the Ring-Down pipeline.
After all physical parameters have been calculated in RD4, RD5 and Rd6, this step collects them and store the result.

The following values are gathered:

- Resonance frequency f_0
- Decay constant τ
- Quality factor $Q = \pi f_0 \tau$
- Damping factor $\alpha = {}^{1}/_{\tau}$
- Linewidth values Δf_τ and Δf_Q
- Fit error (RMSE)

RD7 does not perform new physical calculations.
Its role is to validate the results and save them into a CSV file for later comparison between different runs.
This allows:

- Monitoring stability over time

- Comparing SIM mode and real hardware measurements
- Tracking changes in damping under different electrolyte conditions

RD7 therefore completes the Ring-Down analysis and prepares the data for experimental evaluation.

4 Results and Analysis

This chapter presents the experimental observations obtained during the evaluation of the QCM measurement system. The aim of these experiments is to observe the behaviour of the sensor response and to assess the feasibility of the implemented signal-processing pipelines.

The results include electrochemical measurements obtained through cyclic voltammetry, as well as observation related to frequency response and dissipation behaviour in the QCM system. In addition, the behaviour of the implemented MLA and ring-down pipelines is examined using the acquired data.

The focus if this chapter is on identifying signal characteristics, stability, and observable trends rather than extracting highly optimised physical parameters. This is consistent with the objective of this thesis, which is to evaluate the feasibility and practical behaviour of fast QCM measurement methods using the developed FPGA-based platform.

4.1 Experimental Data Source

The electrochemical data analysed in this work was obtained from laboratory measurements performed using a commercial QCM measurement system. These measurements were conducted in collaboration with Viktor in order to observe electrochemical behaviour in a battery electrolyte environment.

The experiment consisted of cyclic voltammetry (CV) measurements in which the electrode potential was swept across a defined voltage range while the resulting current response was recorded. Cyclic voltammetry is a commonly used electrochemical technique for analysing reactions occurring at electrode–electrolyte interfaces.

The commercial instrument used for these measurements represents a high-precision laboratory system capable of detecting small electrochemical changes and interfacial processes. Such systems are typically expensive and are mainly used in controlled laboratory environments.

In contrast, the goal of this thesis is to investigate the feasibility of developing a significantly lower-cost QCM measurement platform based on FPGA-based signal acquisition and digital signal processing techniques.

4.1.1 Cyclic Voltammetry Measurements

The cyclic voltammetry (CV) measurements were analysed by examining the applied voltage and the corresponding current response over time for multiple scans, as shown in Figure 4.1.

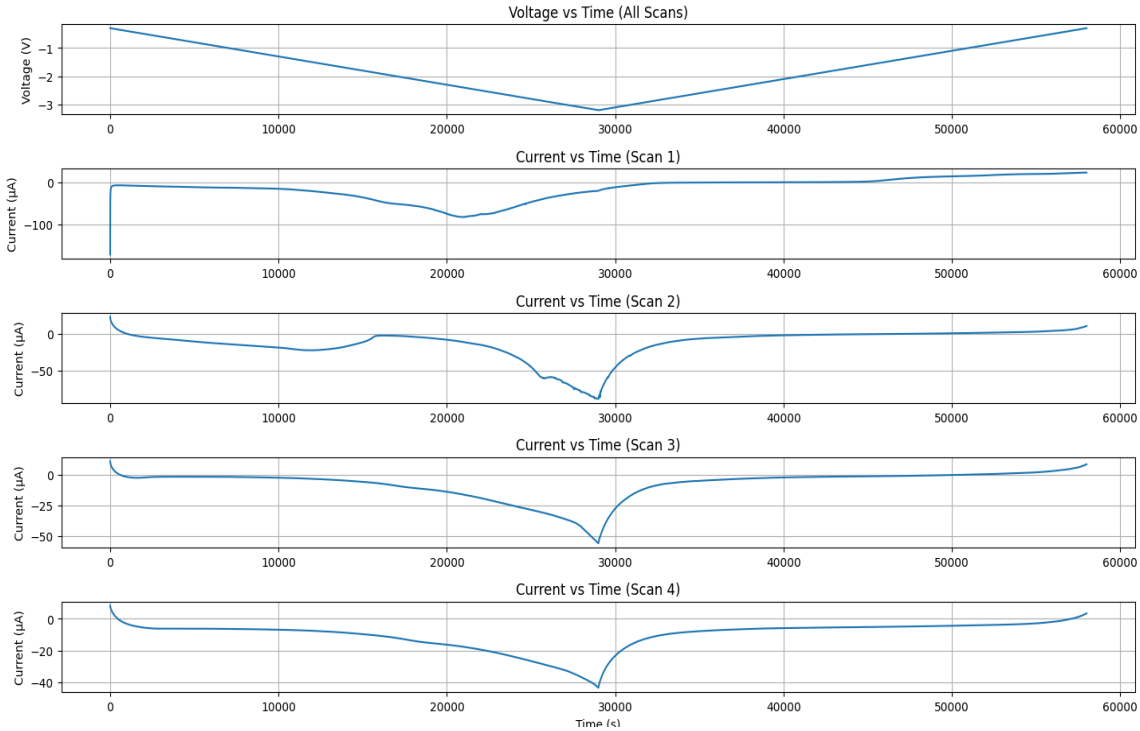


Figure 4.1:Applied voltage and corresponding current response as a function of time for four cyclic voltammetry scans. The follows a consistent triangular waveform in all scans, while the current response varies, showing negative peaks that decrease in magnitude across successive scans.

The applied voltage follows a consistent triangular waveform in all scans, indicating that the excitation signal was stable and repeatable throughout the experiment.

In contrast, the current response varies between scans. During the part of the sweep where the voltage reaches more negative values, the current drops below zero. This indicates that electrochemical reduction reactions are occurring at the electrode-electrolyte interface.

It can be observed that the magnitude of the negative current decreases in the later scans. The first scan shows the strongest response, while the subsequent scans become progressively less negative and smoother. This suggests that the system is changing during repeated measurement.

A possible explanation for this behaviour is the formation of reaction products at the electrode surface, which can influence the electrochemical response over time.

Overall, while the applied voltage remains unchanged, the current response evolves across the scans, indicating changes in the electrochemical system during the measurements.

4.2 QCM Frequency and Dissipation Measurements

The quartz crystal microbalance with dissipation monitoring (QCM-D) technique measures changes in both the resonance frequency and the energy dissipation of an oscillating quartz crystal. These parameters provide information about interactions occurring at the crystal surface when it is exposed to different environments.

In QCM measurements, a decrease in resonance frequency indicates an increase in effective mass or interaction at the crystal surface, while dissipation reflects the energy loss during oscillation and provides information about the viscoelastic properties of the interacting layer.

The measurements analysed in this work include frequency and dissipation responses recorded at multiple overtones of the crystal resonance frequency, specifically the 3rd, 5th, and 7th overtones. Analysing multiple overtones helps to identify differences in the behaviour of the interacting layer and provides additional insight into the response of the QCM sensor in the electrolyte environment.

4.2.1 Frequency and Dissipation Analysis

Figure 4.2 shows the change in frequency shift (Δf) and dissipation shift (ΔD) of the QCM sensor during the cyclic voltammetry experiment for the 3rd, 5th, and 7th overtones, plotted as a function of time.

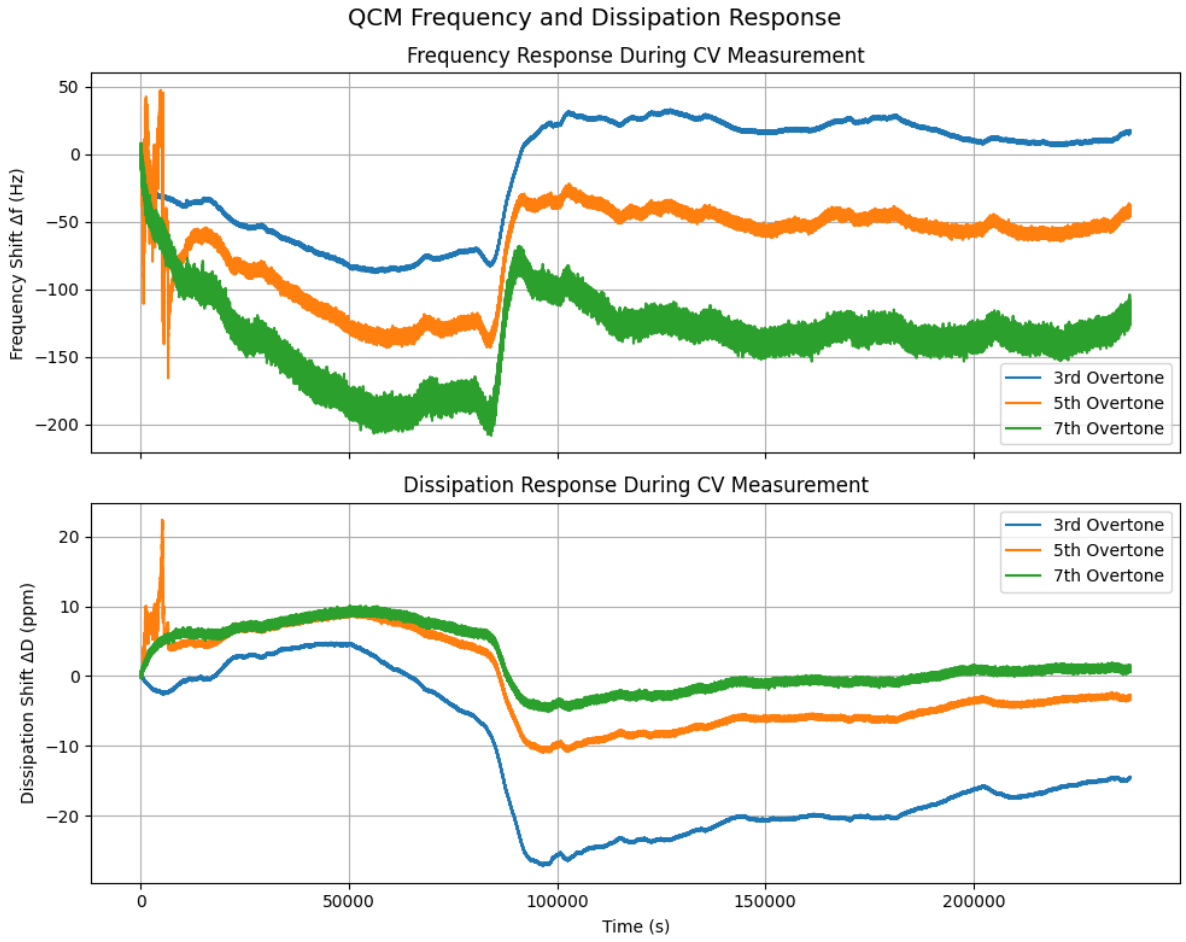


Figure 4.2:Frequency shift (Δf) and dissipation shift (ΔD) of the QCM sensor during cyclic voltammetry measurements, plotted over time for the 3rd, 5th, and 7th overtones. The figure highlights the relationship between electrochemical activity and the corresponding QCM response.

At the beginning of the measurement, both frequency and dissipation signals show initial fluctuations, which are likely associated with system stabilisation. As the experiment progresses, a consistent trend is observed in which the frequency decreases while the dissipation exhibits noticeable variation.

A pronounced transition occurs approximately in the range of 80,000 to 100,000 s, where all three overtones show a clear change. The frequency exhibits a significant drop, while the dissipation shows a corresponding variation within the same time region. This behaviour is consistent across all overtones, with higher overtones showing a larger magnitude of change.

When compared with the cyclic voltammetry results, this transition corresponds to the region where the current becomes strongly negative, indicating the occurrence of electrochemical reactions. These reactions can lead to the formation of decomposition products in the electrolyte, which may alter its physical properties such as viscosity and density.

These changes are reflected in the QCM measurements. The decrease in frequency indicates an increase in effective mass or interaction at the crystal surface, while the change in dissipation suggests variation in the viscoelastic properties of the surrounding medium.

Following this transition, both frequency and dissipation signals gradually stabilise, showing smaller variations over time. This indicates that the system reaches a relatively stable condition after the initial electrochemical processes.

Overall, the results demonstrate a clear relationship between the electrochemical behaviour observed in the CV measurements and the corresponding QCM response.

4.3 MLA and Ring-Down Results

The behaviour of the implemented MLA pipeline was analysed using both time-domain and frequency-domain representations of the measured signal. The time-domain signal (TD0), shown in Figure 3.1 and Figure 3.2, confirms that oscillations are present in the acquired waveform. However, the signal is not a clean sinusoidal waveform and contains noticeable noise. This indicates that, although the QCM excitation is achieved, the measured signal is affected by noise and acquisition limitations.

Before extracting the final response, intermediate validation steps (A3–A8) were performed to verify signal quality. These checks confirmed that a dominant frequency component is present and that the signal can be processed further. However, variations and noise are observed in the signal, which affect the overall stability and quality of the measurement.

The frequency-domain analysis (A9), shown in Figure 3.8, provides the magnitude spectrum, phase response, and an impedance-like representation. A resonance region can be identified, indicating that the dominant oscillation frequency is captured. However, the magnitude spectrum is not smooth and shows fluctuations, and the phase response contains irregular

variations instead of a clean continuous trend. This indicates that the signal quality is limited and affected by noise.

The final MLA result obtained using I/Q demodulation (A10), shown in Figure 3.9 and Figure 3.10, presents the amplitude response and impedance-like curve around the resonance region. The results show a general resonance behaviour near the expected frequency, meaning the response follows the typical shape of a resonance curve. In an ideal case, this response is expected to form a smooth and symmetric peak (similar to a bell-shaped curve) in the amplitude plot, and a smooth symmetric dip in the impedance-like representation.

In contrast, the obtained curves are not smooth and exhibit noticeable variations. The peak is not symmetric, and the impedance-like curve does not show a well-defined minimum. These deviations indicate that, although the MLA method is able to capture the resonance region, the extracted response is affected by noise and limitations in the measurement and processing chain.

In contrast to the MLA method, the ring-down approach did not produce reliable results under the given experimental conditions. The recorded signal did not show a clear exponential decay behaviour and was instead dominated by noise. As a result, it was not possible to extract decay-related parameters such as damping or quality factor from the measured data.

To support the evaluation of the ring-down method, simulated signals were analysed, where a clear exponential decay behaviour was observed. This confirms that the method itself is valid in principle, but its practical implementation is highly sensitive to signal quality and excitation conditions. In this work, these conditions were not sufficiently controlled to obtain meaningful ring-down measurements.

Overall, the MLA method demonstrates the ability to detect resonance behaviour, although with limited accuracy and stability, while the ring-down method was not successfully realised in the experimental setup.

4.4 Relation to the FPGA- Based Measurement Approach

The experimental measurements presented in this chapter were obtained using a commercial QCM-D instrument and illustrate the typical behaviour of a quartz crystal sensor in an electrolyte environment. The observed frequency shifts and dissipation changes reflect the physical interactions occurring at the crystal surface during electrochemical processes.

In this thesis, the primary focus was the development of signal-processing pipelines for an FPGA-based measurement system. These pipelines are intended to analyse key signal characteristics such as resonance frequency, signal stability, and decay behaviour, which are fundamental to QCM measurements.

The experimental results therefore serve as a reference for the type of signals that the FPGA-based system is expected to acquire and process. The frequency and dissipation measurements demonstrate how changes at the sensor surface are reflected in measurable signal variations over time.

When comparing these reference measurements with the results obtained from the implemented MLA and ring-down pipelines, it can be observed that the FPGA-based approach is able to capture the general behaviour of the QCM response. In particular, the MLA method is capable of identifying the resonance region, although the extracted response shows limitations in terms of smoothness and stability. The ring-down method, in its current implementation, did not provide reliable results. The recorded signal was dominated by noise and did not exhibit a clear exponential decay behaviour, making it difficult to extract meaningful decay-related parameters. In addition, limitations related to the FPGA implementation, including the use of the available signal acquisition wrapper, affected the quality and consistency of the recorded signal. These factors suggest that the observed limitations are not solely due to the method itself, but are also influenced by the hardware setup and integration within the FPGA framework.

These observations indicate that, while the FPGA-based framework is able to detect relevant signal features, further improvements are required in signal acquisition, synchronization, and noise reduction to achieve more accurate and stable measurements comparable to established laboratory systems.

5 Discussion

The measurements presented in Chapter 4 illustrate the typical response of a quartz crystal microbalance sensor operating in an electrolyte environment relevant to lithium-ion battery systems. The observed changes in resonance frequency and dissipation reflect interactions occurring at the sensor surface when the crystal is exposed to the electrolyte. These signals were obtained using a commercial QCM-D instrument and therefore represent realistic data that are commonly observed in electrochemical QCM experiments.

Commercial QCM-D systems provide highly accurate and stable measurements, However, they require complex instrumentation and controlled laboratory environments. In addition, such experiments can take several hours or even days to complete in order to observe gradual changes in the sensor behaviour. If experimental errors occur, measurements often need to be repeated, increasing both the time and cost associated with these studies.

The motivation of this thesis is therefore to explore the possibility of analysing QCM sensor signals using a more compact and cost-efficient measurement platform. Instead of relying entirely on large commercial systems, the work focuses on developing signal-processing pipelines that could analyse QCM signals using an FPGA-based platform.

The experimental data obtained from the commercial QCM system serve as a reference for the signals that the proposed FPGA-based system is expected to analyse. By examining the behaviour of frequency and dissipation responses across multiple overtones, it becomes possible to identify the signal characteristics required for the reliable QCM analysis.

The results obtained from the FPGA-based implementation show that the MLA pipeline is able to detect the presence of a resonance region. The extracted amplitude and impedance-like responses follow the general shape of a resonance curve; however, the results are affected by noise and lack smoothness and symmetry compared to ideal behaviour. This indicates that, while the approach is functional, the measurement quality is currently limited.

In contrast, the ring-down method did not produce reliable results under the given experimental conditions. The recorded signals were dominated by noise and did not exhibit a clear decay

behaviour. In addition, limitations in the FPGA-based acquisition setup, including the signal interface and wrapper implementation, affected the ability to capture a stable transient response. This suggests that the limitations are not due to the method itself, but rather due to the current hardware and integration constraints.

Overall, the study demonstrates that FPGA-based signal-processing pipelines can capture key features of QCM signals, but further improvements in signal acquisition, noise reduction, and system integration are required to achieve stable and accurate measurements.

6 Conclusion

This thesis investigated the feasibility of analysing quartz crystal microbalance (QCM) senor signals using a compact FPGA-based measurement platform. The motivation for this work arises from the complexity and high cost of commercial QCM-D instruments, which require specialised laboratory setups and long experimental measurement times.

In this work, two signal-processing pipelines were developed and evaluated using an FPGA-based Red Pitaya platform. The MLA pipeline demonstrated that the QCM sensor can be excited and that resonance-related signal components can be detected using the FPGA-based platform. However, the extracted results were affected by noise and lacked the smoothness required for reliable quantitative analysis, indicating that further improvements in signal quality and system design are necessary.

The ring-down pipeline explored the possibility of analysing the transient decay behaviour of the QCM signal after excitation. However, the experimental implementation did not produce a clear decay response, and the recorded signals were dominated by noise. As a result, it was not possible to reliably extract decay-related parameters from the measured data. This limitation is attributed to constraints in the signal acquisition and FPGA-based implementation rather than the method itself.

Overall, this work demonstrates that FPGA-based systems can be used to detect and analyse key characteristics of QCM signals, providing a proof-of-concept for a more compact and cost-effective measurement approach. However, further development is required to improve measurement stability and accuracy before such systems can be used as an alternative to commercial QCM-D instruments.

7 Future Work

Future work should focus on improving the excitation control, optimizing the analog front-end design, and enhancing signal acquisition quality within the FPGA framework. In addition, better synchronization between signal generation and acquisition is required, particularly for reliable implementation of the ring-down method.

Further development of noise reduction techniques and improved signal conditioning could significantly enhance the quality of the extracted resonance response. Integrating the QCM sensor more closely with the FPGA-based platform may also enable more compact and flexible measurement systems.

These improvements could support future applications where QCM sensors are directly integrated into electrochemical systems, such as battery monitoring and in-situ analysis.

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